INCH-POUND

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MILITARY SPECIFICATION

TACHOMETER GENERATOR, AC; GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for tachometer generators. It is not complete in itself, but should be used in conjunction with MIL-S-81963 which will form an inherent part of this specification.

1.2 Classification.

1.2.1 Nomenclature. The nomenclature consists of the item name, Tachometer Generator, followed by a type designation and an uppercase modification letter. All tachometer generators having the same nomenclature should be electrically and mechanically interchangeable for all military applications. The type designation is made up of a combination of digits and letters as explained in the following paragraphs. Types other than 115 volts should be prefixed by applicable voltage identification followed by the letter "V", such as "26V". The type designation of a 15IH4A is illustrated in Table I. Nomenclature for new or modified tachometer generator types will be assigned by the Naval Air Systems Command, Department of the Navy.

1.2.1.1 Size. The first two digits designate the maximum diameter in tenths of an inch. If the diameter is not exactly a whole number of tenths, the next higher tenth is used.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.2 Function. The succeeding letter group indicates one of the following functions:

RG	Rate tachometer generator
IH	Integrating tachometer generator, heater compensated
IN	Integrating tachometer generator, network compensated
DG	Damping tachometer generator

1.2.1.3 Supply frequency. The succeeding digit(s) identifies the frequency of the power supply in accordance with the following code:

Supply frequency (Hz)	Code
60	6
400	4

New codes associated with supply frequencies not indicated will be assigned as required by the Naval Air Systems Command, Department of the Navy.

1.2.1.4 Modification. An upper case letter "A" following the frequency digit indicates the basic issue of a standard tachometer generator. The first modification that significantly affects the external dimensions or the electrical characteristics of the basic type should be indicated by the upper case letter "B". Succeeding modifications will be indicated by "C", "D", etc., except the use of letters "I", "L", "O", and "Q" is prohibited.

1.2.2 Illustration. A tachometer generator classified as a 15IH4A (see Table I) indicates the original design of a 115 volt, 400 Hz, heater-compensated, integrating tachometer generator whose body is greater than 1.40 inches but not greater than 1.50 inches. If the unit had an input voltage requirement of 26 volts, the type designation would be 26V-15IH4A.

1.2.3 Part identifying number. The part identifying number (PIN) consists of the letter "M", the basic number of the specification sheet (not including the revision letter), an assigned dash number, and an upper case suffix letter designating the latest modification letter in the type designation, as shown in Table II.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are

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those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-13924	Coating, Oxide, Black, for Ferrous Metals
MIL-S-81963	Servocomponents, Precision Instrument, Rotating, Common Requirements and Tests; General Specification for
MIL-T-22821/1	Tachometer Generator, Type 15IH4A
MIL-T-22821/2	Tachometer Generator, Type 15RG4A
MIL-W-16878/4	Wire, Electrical, Polytetraflouroethylene (PTFE), Insulated, 200°C, 600 Volts, Extruded Insulation
MIL-W-16878/6	Wire, Electrical, Polytetraflouroethylene (PTFE), Insulated, 200°C, 250 Volts, Extruded Insulation

STANDARDS

MILITARY

MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods
MS35276	Screw, Machine-Drilled Fillister Head, Slotted Corrosion Resisting Steed, Passivated, UNF-2A
MS35338	Washer, Lock-Spring, Helical, Regular (Medium) Series

(Unless otherwise indicated, copies of the referenced federal and military specifications, standards, and handbooks, are available from the Naval Publications and Forms Center (ATTN: Documents Order Desk), Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

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AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 207.06	Tooth Proportions for Fine Pitch Involute Spur and Helical Gears
AGMA 2000-A	Gear Classification and Inspection Handbook Tolerances and Measuring Methods for Unassembled Spur and Helical Gears (Including Metric Equivalents)

(Application for copies should be addressed to the American Gear Manufacturers Association, 1500 King Street, Suite 201, Alexandria, VA 22314.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents may also be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein, in MIL-S-81963, and in accordance with the applicable specification sheets. In the event of any conflict between requirements of MIL-S-81963, this specification, and the specification sheet, the latter shall govern.

3.2 First article. When specified in the contract, a sample shall be subjected to first article inspection (see 4.6 and 6.3).

3.3 Quality conformance. The tachometer generators furnished under this specification shall be inspected and pass the quality conformance tests specified herein (see 4.7).

3.4 Parts, materials and processes.

3.4.1 Parts and materials. Tachometer generators shall be constructed of parts and materials in accordance with MIL-S-81963. If a suitable material is not listed, the material used shall be such that the tachometer generator shall conform to all the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

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3.4.2 Rotor shafts and housings. Rotor shafts and housings shall meet the requirements of MIL-S-81963 having a hardness equivalent to 28HRC to 35HRC or as specified in the applicable specification sheet. If a flat black finish is desired on the housing, the finish shall be in accordance with MIL-C-13924.

3.4.3 Wire.

3.4.3.1 Magnet wire. Magnet wire shall conform to J-W-1177, Class 105, requirements for the high temperature specified for the tachometer generator, recognizing the addition of the self-heating of the unit. Type and thickness of insulation shall be left to the discretion of the manufacturer to meet the temperature requirements of the unit.

3.4.3.2 Lead wire. Wire leads shall consist of seven strands of silver-coated copper conductor and shall be a minimum of 18 inches long. For servocomponent frame size 05, wire leads shall conform to MIL-W-16878/6, Type ET-30. For servocomponent frame sizes 08, 11 and 15, wire leads shall conform to MIL-W-16878/4, Type E-28. Wire leads shall then meet the requirements of MIL-S-81963.

3.5 Design and construction.

3.5.1 Dimensions. The physical dimensions of the tachometer generator shall be identified by the letters shown on the outline drawing included as Figure 1. The lettered dimensions are provided in Table III.

3.5.2 Terminal and lead wire identification. Screw or solder type terminations of the tachometer generator section and servomotor section shall be as specified in MIL-S-81963. When lead wires are specified, the wire shall conform to 3.4.3.2. Standard lead wire and terminal designation shall be color coded as specified in Figure 2 and Table IV. Terminal hardware is listed in Table V.

3.5.3 Direction of rotation. The standard positive direction of rotation is counterclockwise.

3.5.4 Spur gear data. Spur gears shall be designed in accordance with AGMA 207.06 and tested in accordance with AGMA 2000-A. Tooth form shall be full depth involute.

3.5.5 Storage. Tachometer generators shall be capable of storage in ambient temperatures ranging from 0 to 60 degrees Centigrade.

3.6 Performance.

3.6.1 Visual and mechanical inspection. Visual and mechanical inspection shall be performed in accordance with 4.8.1.

3.6.2 Mechanical breakaway torque. When tested in accordance with 4.8.2, the maximum bearing friction torque of the tachometer generator shall not exceed the value specified in the applicable specification sheet.

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3.6.3 Shaft radial and end play. When tested in accordance with 4.8.3, the radial and end play of the tachometer generator shall not exceed the values specified in the applicable specification sheet.

3.6.4 Total runout. When tested in accordance with 4.8.4, the total runout of the smooth portion of the shaft shall not exceed the value specified in the applicable specification sheet.

3.6.5 Dielectric withstanding voltage. When tested in accordance with 4.8.5 and Table VI, the tachometer generator shall meet the requirements of MIL-S-81963.

3.6.6 Insulation resistance. When tested in accordance with 4.8.6 and Table VI, insulation resistance measurements shall meet the requirements of MIL-S-81963.

3.6.7 Current and power, heater windings. When tested in accordance with 4.8.7, the maximum value of the current drawn and the maximum value of the power consumed in the heater winding, shall not exceed those values specified in the applicable specification sheet.

3.6.8 Current. When tested in accordance with 4.8.8, the current drawn shall be within the limits specified in the applicable specification sheet.

3.6.9 Power. When tested in accordance with 4.8.9, the power consumed shall be within the limits specified in the applicable specification sheet.

3.6.10 Direction of rotation and polarity. When tested in accordance with 4.8.10, the output voltage at terminal two shall be approximately 180 degrees out of phase with the voltage at terminal one of the energizing winding when the rotor is driven in a counterclockwise direction. Direction is determined when viewing the unit from the shaft extension end associated with the mounting surfaces.

3.6.11 Axis error voltages. When tested in accordance with 4.8.11, the axis error voltages measured at any speed in the specified speed range shall be no greater than those values specified in the applicable specification sheet (see 6.6.5.4).

3.6.12 Zero-speed output voltages. Zero speed output voltages which may be required include total rms null, fundamental rms null, fundamental in-phase null and fundamental quadrature-phase null. When tested in accordance with 4.8.12, the zero-speed output voltages shall be no greater than the values specified in the applicable specification sheet (see 6.6.5.2).

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3.6.13 Position error voltages. When tested in accordance with 4.8.13, the position error voltages shall meet the requirements of the applicable specification sheet (see 6.6.5.3).

3.6.14 In-phase speed-sensitive transformation ratio. When tested in accordance with 4.8.14, the in-phase speed-sensitive transformation ratio shall be within the limits specified for the calibration speed indicated in the applicable specification sheet.

3.6.15 Phase shift. Phase shift shall be tested in accordance with 4.8.15, and shall be no greater than the requirements in the applicable specification sheet, for any speed in the specified speed range.

3.6.16 Warm-up time. The time required for the measured in-phase speed-sensitive transformation ratio and the in-phase axis error voltage to be within the limits specified in the applicable specification sheet and subsequently remain within those limits is the warm-up time. When tested in accordance with 4.8.16, the warm-up time shall conform to the requirements of the applicable specification sheet.

3.6.17 Impedance. The impedance shall be within the limits specified in the applicable specification sheet when tested in accordance with 4.8.17.

3.6.18 In-phase linearity. When tested in accordance with 4.8.18, the percent of deviation from the in-phase linearity at any speed in the specified speed range shall be no greater than the value required in the applicable specification sheet. Unless otherwise specified, the calibration speed (Nc) shall be 3,000 rpm.

3.6.19 Voltage sensitivity. When tested in accordance with 4.8.19, the unit shall meet the requirements of Table VII and the applicable specification sheet.

3.6.20 Frequency sensitivity. When tested in accordance with 4.8.20, tachometer generator shall meet the requirements of Table VII and the applicable specification sheet.

3.6.21 Temperature rise. When measured in accordance with 4.8.21, the temperature rise shall be no greater than the value specified in the applicable specification sheet.

3.6.22 Temperature sensitivity. When tested in accordance with 4.8.22, the tachometer generator shall meet the requirements of Table VII and the applicable specification sheet.

3.6.23 Rotor moment-of-inertia. The rotor moment-of-inertia shall be tested in accordance with 4.8.23 and shall meet the requirements of the applicable specification sheet.

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3.6.24 Security of terminals or wire leads. When tested in accordance with 4.8.24, terminals or wire leads shall meet the requirements of MIL-S-81963.

3.7 Environmental performance.

3.7.1 Vibration. Tachometer generators shall meet the requirements of MIL-S-81963 and Table VII herein after vibration is conducted in accordance with 4.9.1.

3.7.2 Shock.

3.7.2.1 Shock, low impact. Tachometer generators shall meet the requirements of MIL-S-81963 and Table VII herein after low shock is conducted in accordance with 4.9.2.1.

3.7.2.2 Shock, high impact. The tachometer generators shall meet the requirements of MIL-S-81963 and Table VII herein after high shock is conducted in accordance with 4.9.2.2.

3.7.3 Altitude. Tachometer generators shall be capable of operation from sea level to 100,000 feet in combination with any ambient temperature from -55°C to the high ambient temperature specified in the applicable specification.

3.7.3.1 Altitude, low temperature. The tachometer generator shall be tested in accordance with 4.9.3.1 and shall meet the requirements of MIL-S-81963 and Table VII herein.

3.7.3.2 Altitude, high temperature. The tachometer generator shall be tested in accordance with 4.9.3.2 and shall meet the requirements of MIL-S-81963 and Table VII herein.

3.7.4 Ambient temperature. Tachometer generators shall be capable of operation in ambient temperatures ranging from -55°C to the high ambient temperature specified in the applicable specification sheet.

3.7.4.1 Ambient, low temperature. When tested in accordance with 4.9.4.1, the tachometer generator shall meet the requirements of MIL-S-81963 and Table VII herein.

3.7.4.2 Ambient, high temperature. When tested in accordance with 4.9.4.2, the tachometer generator shall meet the requirements of MIL-S-81963 and Table VII herein.

3.7.5 Endurance. Endurance shall be conducted in accordance with 4.9.5 and Table VIII. After testing, the tachometer generators shall meet the requirements of MIL-S-81963 and Table VII herein.

3.7.6 Moisture resistance. Tachometer generators shall be capable of operation and storage in climates of high relative humidity. After

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testing in accordance with 4.9.6, the tachometer generators shall conform to the requirements of MIL-S-81963 and Table VII.

3.7.7 Audible noise, structureborne. When required by the applicable specification sheet, structureborne noise shall be tested in accordance with 4.9.7 and shall not exceed the limiting value specified in the applicable specification sheet.

3.7.8 Salt atmosphere resistance. The tachometer generator shall be capable of withstanding atmospheres saturated with salt laden moisture. If required by the applicable specification sheet, the tachometer generator shall be tested in accordance with 4.9.8, disassembled and visually inspected for evidence of destructive deterioration, and shall meet the requirements of MIL-S-81963.

3.7.9 Explosion resistance. When required by the applicable specification sheet, tachometer generators shall meet the requirements of MIL-S-81963 when tested in accordance with 4.9.9.

3.8 Identification marking. Identification markings shall be as specified in MIL-S-81963.

3.9 Workmanship. Workmanship of the tachometer generator shall conform to the requirements of MIL-S-81963.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Responsibility for inspection shall be as specified in MIL-S-81963.

4.1.1 Responsibility for compliance. The responsibility for compliance with all requirements of sections 3 and 5 shall be in accordance with MIL-S-81963.

4.2 Test equipment and inspection facilities. The accuracy of test equipment and inspection facilities shall conform to MIL-S-81963.

4.3 Test conditions.

4.3.1 Standard test conditions. Unless otherwise specified, the standard test conditions shall be as specified in MIL-S-81963, and each test shall be carried out with the tachometer generator mounted in the test fixture illustrated in Figure 3.

4.3.2 Ambient test conditions. Unless otherwise specified herein, all measurements shall be made within the following temperature, atmospheric pressure and relative humidity limits.

Temperature - 23° ± 2°C
Pressure - 28 to 32 inches Hg
Humidity - No greater than 55 percent

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4.3.3 Standard test voltages. Unless otherwise specified, the standard test voltage sources shall possess the characteristics of 4.3.3.1 - 4.3.3.3.

4.3.3.1 Amplitude. The amplitude of the standard test voltage for the tachometer generator shall be the rated or specified value ± 1 percent. The standard test voltage shall be derived from a source capable of being maintained within 0.1 percent for damping types over the length of time the test requires. For integrating types, the source shall be maintained within .01 percent of the required value.

4.3.3.2 Frequency. The frequency of the standard test voltage applied to the tachometer generator shall be within 0.1 percent of the rated or specified value for damping types, and within 0.01 percent of rated or specified value for integrating types.

4.3.3.3 Total harmonic distortion. The total harmonic content of the standard test voltage applied to the tachometer generator shall be less than ± 1 percent for damping types, and less than 0.1 percent for computing types.

4.3.4 Temperature, stabilized operating. The stabilized operating temperature of a tachometer generator shall be as specified in MIL-S-81963, The rated voltages specified in the applicable specification sheet shall be applied to the energizing winding (and heater winding if applicable) of the tachometer generator. The output winding shall be used for the periodic DC resistance measurement.

4.3.5 Temperature, stabilized non-operating. The stabilized non-operating temperature of a tachometer generator shall be as specified in MIL-S-81963, using the output winding for the periodic DC resistance measurement.

4.3.6 Standard load. The standard load shall be as specified in the applicable specification sheet.

4.4 Test equipment.

4.4.1 Speed-sensitive output measuring equipment. The required speed-sensitive transformation ratios, phase shift angles, and axis error of tachometer generators shall be measured according to Figure 4, or by a method which produces equivalent results. For integrating tachometer generators, the input capacitance of the output measuring device shall be no greater than 30 pF.

4.4.2 Voltage measuring equipment. The method used to determine the transformation ratio of the tachometer generator shall be a ratio device in conjunction with a phase sensitive null detector. The quadrature-phase output should also be determined as the transformation ratio. The quadrature-phase reference of the phase sensitive detector used shall be in

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quadrature with the in-phase reference, ± 10 minutes. The ratio device employed shall be linear within 0.001 percent over the range of nulling voltages required by the generator output. The voltmeter used to determine the magnitudes of the required voltages shall be of the type best described as a "linear average indicating type" calibrated within ± 2 percent of full scale to indicate the rms value of a pure sine wave. It shall also possess the characteristic, when required, to attenuate signals other than the carrier at least 100 times (40 dB). The input impedance of the voltage measuring equipment shall be compatible with the standard load specified in the applicable specification sheet.

4.4.3 Driving speed equipment. Unless otherwise specified in the applicable specification sheet, the equipment employed to determine the speed-sensitive outputs of the tachometer generator shall be capable of driving the shaft of the tachometer generator at the specified speeds within 0.003 percent.

4.5 Classification of tests. The method of sampling, inspections and tests performed on tachometer generators shall be classified as follows:

a. First article (4.6)

b. Quality conformance (4.7)

4.5.1 Degradation of performance. The following minor relaxations may be permitted at the discretion of the qualifying activity.

4.5.1.1 Quality conformance tests. All the specified performance characteristics must be attained.

4.5.1.2 Environmental tests. As appropriate during or following each of the environmental tests, the following minor relaxations in specified requirements may be permitted. It should be noted that the relaxations are not cumulative; for example, axis-error voltages, if accepted at 95 percent of the specified minimum value following vibration, must still be not less than 95 percent of the specified minimum value following low shock.

4.5.1.3 Shaft radial play. After high impact shock, the radial play shall be no greater than 1.5 times the value specified in the applicable specification sheet.

4.5.1.4 Shaft end play. After high impact shock, the end play shall be no greater than 1.5 times the value specified in the applicable specification sheet.

4.5.1.5 Mechanical breakaway torque. Following high impact shock, an increase to 3 times the specified maximum value is permitted.

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4.5.1.6 Dielectric withstanding voltage. Following high impact shock, a winding leakage current increase of 1.5 mA peak maximum is permitted.

4.5.1.7 Insulation resistance. Following high impact shock and having been immediately preceded by dielectric withstanding voltage, a reduction to 25 megohms insulation resistance is permitted.

4.5.1.8 Axis-error voltages, zero-speed output voltages, position-error voltages, in-phase speed-sensitive transformation ratio. A 5 percent increase or reduction in the specified values is permitted. Following high impact shock, a ± 20 percent variation of the in-phase speed-sensitive transformation ratio is allowed.

4.5.1.9 Major failures during or following environmental tests. Allowances having been made for the relaxations quoted above, failures experienced during or following environmental tests shall be cause for refusal to grant first article approval.

4.6 First article sample inspection. When required by the contracting activity, first article shall be as specified in MIL-S-81963.

4.6.1 First article sample failure. Action following first article sample failure shall be as specified in MIL-S-81963.

4.7 Quality conformance inspection. Quality conformance inspection shall be as specified in MIL-S-81963 and Table VII herein.

4.7.1 Quality conformance inspection sampling. Sampling for acceptance inspection shall be as specified in MIL-S-81963.

4.7.2 Quality conformance inspection routine. Quality conformance inspection routine shall be as specified in MIL-S-81963 and Table VII herein.

4.7.3 Quality conformance sample failure. Action following quality conformance sample failure shall be as specified in MIL-S-81963.

4.8 Test methods and examinations.

4.8.1 Visual and mechanical inspection. The tachometer generator shall be carefully inspected in accordance with MIL-S-81963 and shall meet the requirements of 3.6.1 and the applicable specification sheet.

4.8.2 Mechanical breakaway torque. Mechanical breakaway torque shall be performed in accordance with MIL-S-81963 with the frame of the generator being rotated at two to three rpm. The tachometer generator shall meet the requirements of 3.6.2. For first article, this test shall also be performed at -55°C and the high ambient temperature specified in the applicable specification sheet.

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4.8.3 Shaft radial and end play. Shaft radial and end play shall be tested in accordance with MIL-S-81963 and shall meet the requirements of 3.6.3.

4.8.4 Total runout. Total runout of the smooth portion of the tachometer generator shaft shall be measured in accordance with MIL-S-81963. Total runout shall not exceed the requirements of 3.6.4.

4.8.5 Dielectric withstanding voltage. A 60 Hz test potential of the magnitude specified by Table VI shall be applied to the indicated test points. The test potential shall be raised slowly (a minimum time of 3 seconds) to the specified value and held at the value for one minute; then reduced slowly (minimum time, 3 seconds) to zero. Peak value of the test potential throughout this test shall not exceed 1.5 times the specified test potential. Failure shall be indicated by a winding leakage current in excess of one peak milliampere or the indication of internal arcing or corona discharge. The failure leakage current limit of one peak milliampere shall not include that current drawn by the equipment capacitance or the capacitance of the windings under test. Test equipment employed shall be such as to differentiate between winding leakage current and surge discharge current. If the tachometer generator successfully meets the requirements of 3.6.5, insulation resistance shall be performed immediately.

4.8.6 Insulation resistance. The DC resistance between the terminal locations specified in Table VI shall be measured in accordance with MIL-S-81963. A minimum of 500 volts DC shall be applied between all points specified in Table VI. Tachometer generators shall then meet the requirements of 3.6.6. Quality conformance tests shall be conducted at the standard ambient temperature of 4.3.2. First article tests shall be conducted at -55°C and the normal operating temperature, ± 5°C, specified in the applicable specification sheet.

4.8.7 Current and power, heater winding. The tachometer generator shall be mounted in the standard test fixture and allowed to remain unexcited for three hours at the standard test conditions of 4.3.1. The voltage of proper amplitude and frequency specified in the applicable specification sheet shall then be applied to the heater winding or windings. The total current drawn and the total power consumed in the heater circuit shall conform to the requirements of 3.6.7.

4.8.8 Current. The tachometer generator shall be mounted on the test fixture of Figure 3 and the output winding loaded with the load specified in the applicable specification sheet. The tachometer generator shall then be brought to the stabilized operating temperature of 4.3.4 and the current drawn by the energizing winding shall be measured and shall meet the requirements of 3.6.8.

4.8.9 Power. The tachometer generator shall be mounted on the test fixture of Figure 3 and the output winding loaded with the standard load specified in the applicable specification sheet. The tachometer generator

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shall then be brought to the stabilized operating temperature of 4.3.4 and the power consumed by the energizing winding shall be measured and shall conform to the requirements of 3.6.9.

4.8.10 Direction of rotation and polarity. The output winding of the tachometer generator shall be loaded with the standard test load specified in 4.3.6, and excited in accordance with Figure 2, with the rated standard test voltage applied to the energizing voltage winding. The shaft shall be driven at the speeds indicated in the applicable specification sheet in a counterclockwise (positive) direction. With a suitable detector, the output voltage at terminal two shall be approximately 180 degrees out of phase with the fundamental voltage at terminal one, as required by 3.6.10.

4.8.11 Axis error voltages. The tachometer generator shall be mounted on the test fixture of Figure 3 and the output winding loaded with the standard test load specified in the applicable specification sheet. The tachometer generator shall be brought to the stabilized operating temperature of 4.3.4. At the test speed specified, the in-phase and the quadrature-phase axis error voltages shall be measured and meet the requirements of 3.6.11.

4.8.12 Zero-speed output voltages. The tachometer generator shall be mounted on the test fixture of Figure 3, the output winding loaded with the standard test load specified in 4.3.6, and brought to the stabilized operating temperature of 4.3.4. At a constant speed no greater than one-half rpm, the zero speed output voltages required shall be measured for one complete revolution of the shaft, and shall conform to the requirements of 3.6.12.

4.8.13 Position error voltages. The tachometer generator shall be mounted on the test fixture of Figure 3 and the output winding loaded with the standard test load specified in 4.3.5. The tachometer generator shall then be brought to the stabilized operating temperature of 4.3.4. The in-phase and quadrature-phase position error voltages shall be measured at a constant speed no greater than one-half rpm and shall meet the requirements of 3.6.13 (see Figure 5).

4.8.14 In-phase speed-sensitive transformation ratio. The tachometer generator shall be mounted on the test fixture of Figure 3, the output winding loaded with the standard load specified in 4.3.6, and the unit brought to the stabilized operating temperature of 4.3.4. At the speeds required in the applicable specification sheet, the in-phase speed-sensitive output voltage shall be measured as illustrated in Figure 4 and the transformation ratio determined. The in-phase speed sensitive transformation ratio as defined in 6.7.5.5.1 shall conform to the requirements of 3.6.14.

4.8.15 Phase shift. The tachometer generator shall be mounted on the test fixture of Figure 3, the output winding loaded with the standard test load specified in 4.3.6, and the unit allowed to attain the stabilized operating temperature of 4.3.4. The in-phase and quadrature-phase speed-

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sensitive output voltages shall be measured as illustrated in Figure 4 and shall meet the requirements of 3.6.15.

4.8.16 Warm-up time. When required by the applicable specification sheet, the tachometer generator shall be mounted on the test fixture of Figure 3, loaded with the standard test load specified in 4.3.6, and allowed to reach the stabilized non-operating temperature of 4.3.5. At the end of three hours, the energizing winding and the heater winding, if applicable, shall be excited with rated standard test voltages. Measure the in-phase speed-sensitive transformation ratio and in-phase axis error voltage at 3,000 rpm unless otherwise specified. Warm-up time shall meet the requirements of 3.6.16.

4.8.17 Impedance. The tachometer generator shall be mounted in the test fixture of Figure 3, the output winding loaded with the standard load specified in 4.3.6, and the unit allowed to attain the stabilized operating temperature. The impedance parameters of the energizing voltage winding excited with the rated standard test voltage shall be measured. With the energizing voltage winding still excited, the impedance parameters of the output winding, without the standard load, shall be measured with a fundamental rms voltage across the winding equivalent to that developed at 3,000 or 3,600 rpm. The method used to measure the impedance shall consist of true rms reading dynamometer ammeters, voltmeters and wattmeters with full scale accuracies of 0.25 percent or better. Other methods may be used which achieve the same results. The impedance shall conform to the requirements of 3.6.17.

4.8.18 In-phase linearity. The tachometer generator shall be mounted on the test fixture of Figure 3, the output winding loaded with the standard load specified in 4.3.6, and the unit allowed to attain the stabilized operating temperature. Unless otherwise specified, the in-phase speed-sensitive output voltage shall be measured as in Figure 4 at the calibration speed of 3,000 rpm. The in-phase speed-sensitive output voltages at the speeds specified in the applicable specification sheet shall also be measured. The percent deviation from linearity, as defined in 6.7.8, shall be calculated at each required speed and shall conform to the requirements of 3.6.18.

4.8.19 Voltage sensitivity. Voltage sensitivity tests for the tachometer generator shall be performed with the unit mounted in the standard test fixture of Figure 3 and at the stabilized operating temperature of 4.3.4.

4.8.19.1 In-phase transformation ratio - voltage sensitivity test. The in-phase speed sensitive transformation ratio at the specified calibration speed, shall be measured as specified in 4.8.14 and shall conform to the requirements of 3.6.19 when the energizing winding voltage is first 110 percent of the rated standard voltage, and then 90 percent of the rated standard value.

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4.8.19.2 Phase shift - voltage sensitivity test. The phase shift at the specified calibration speed shall be measured as in 4.8.15 with the energizing winding voltage first 110 percent of the rated standard voltage, and then 90 percent of the rated standard value. The measurements shall conform to the requirements of 3.6.19.

4.8.19.3 Axis voltage - voltage sensitivity test. The in-phase and quadrature-phase axis voltages shall be measured as in 4.8.11 when the energizing winding voltage is first 110 percent of the rated standard voltage and then at 90 percent of the rated standard value. The measurements shall conform to the requirements of 3.6.19.

4.8.19.4 Position voltage - voltage sensitivity test. The in-phase and quadrature-phase position voltages shall be measured as in 4.8.13 when the energizing winding voltage is first 110 percent of the rated standard test voltage, and then at 90 percent of the rated voltage. The measurements shall conform to the requirements of 3.6.19.

4.8.19.5 Zero-speed voltage - voltage sensitivity test. The fundamental rms null voltage and the total null voltage shall be measured as in 4.8.12 and shall conform to the requirements of 3.6.19 when the energizing winding voltage is first 110 percent of the rated standard voltage, and then 90 percent of the rated standard value.

4.8.20 Frequency sensitivity tests. Frequency sensitivity tests shall be performed with the tachometer generator mounted in the standard test fixture and at the stabilized operating temperature of 4.3.4.

4.8.20.1 In-phase transformation ratio - frequency sensitivity test. The in-phase speed sensitive transformation ratio at the specified calibration speed, shall be measured as specified in 4.8.14 and shall conform to the requirements of 3.6.20 when the frequency of the energizing winding voltage is first 105 percent of the rated standard voltage, and then 95 percent of the rated standard value.

4.8.20 2 Phase shift - frequency sensitivity test. The phase shift at the specified calibration speed shall be measured as in 4.8.15 with the frequency of the energizing winding first 105 percent of the rated standard voltage, and then 95 percent of the rated standard value. The measurements shall conform to the requirements of 3.6.20.

4.8.20.3 Axis voltage - frequency sensitivity test. The in-phase and quadrature-phase axis voltages shall be measured as in 4.8.11 when the frequency of the energizing winding voltage is first 105 percent of the rated standard voltage, and then 95 percent of the rated standard value. The measurements shall conform to the requirements of 3.6.20.

4.8.20.4 Position voltage - frequency sensitivity test. The in-phase and quadrature-phase position voltages shall be measured as in 4.8.13 and shall conform to the requirements of 3.6.20 when the energizing winding voltage frequency is first 105 percent of the rated standard voltage, and then 95 percent of the rated standard value.

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4.8.20.5 Zero-speed voltage - frequency sensitivity test. The fundamental rms null voltage and the total null voltage shall be measured as in 4.8.12 and shall conform to the requirements of 3.6.20 when the frequency of the energizing winding voltage is first 105 percent of the rated standard voltage, and then 95 percent of the rated standard value.

4.8.21 Temperature rise. The tachometer generator shall be mounted on the standard test fixture of Figure 3 and allowed to reach the stabilized non-operating temperature of 4.3.5 in a chamber of three to five cubic feet of space per tachometer generator. The DC resistance of the output winding (with standard load removed) shall be measured. After the tachometer generator has been stabilized in accordance with 4.3.4, the temperature rise shall be determined by the temperature rise formula in MIL-S-81963 and conform to the requirements of 3.6.21.

4.8.22 Temperature sensitivity tests. The required ambient temperature sensitivity tests shall be performed with the tachometer generator mounted on the standard test fixture of Figure 3.

4.8.22.1 In-phase transformation ratio - temperature sensitivity test. The tachometer generator shall remain unexcited for a period of three hours at the low ambient temperature, ± 2°C, associated with the temperature sensitivity requirement in the applicable specification sheet. The generator shall then be excited to reach the stabilized operating temperature of 4.3.4. At this temperature condition the in-phase speed-sensitivity transformation ratio shall be measured as in 4.8.14, at the specified calibration speed, and shall conform to the requirements of 3.6.22. The same test shall be repeated after remaining unexcited for three hours at that high ambient temperature, ± 2°C, associated with the temperature sensitivity requirement in the applicable specification sheet.

4.8.22.2 Phase shift - temperature sensitivity test. The tachometer generator shall remain unexcited for three hours at the low ambient temperature, ± 2°C, associated with the temperature sensitivity requirement in the applicable specification sheet. The generator shall then be excited to reach the stabilized operating temperature of 4.3.4. At this temperature condition, the phase shift at the specified calibration speed shall be measured as in 4.8.15 and shall conform to the requirements of 3.6.22. The same test shall be repeated after remaining unexcited for three hours at that high ambient temperature ± 2°C, associated with temperature sensitivity requirement in the applicable specification sheet.

4.8.22.3 Axis voltage - temperature sensitivity test. The tachometer generator shall remain unexcited for three hours at the low ambient temperature, ± 2°C, associated with the temperature sensitivity requirement in the applicable specification sheet. The tachometer generator shall then be excited to reach the stabilized operating temperature of 4.3.4. At this temperature condition, the in-phase and quadrature-phase axis voltages shall be measured as in 4.8.11 and shall meet the requirements of 3.6.22. The same test shall be repeated after remaining unexcited for three hours at the high ambient temperature, ± 2°C, associated with the temperature

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high ambient temperature, ± 2°C, associated with the temperature sensitivity requirement in the applicable specification sheet.

4.8.22.4 Position voltage - temperature sensitivity test. The tachometer generator shall remain unexcited for three hours at the low ambient temperature, ± 2°C, associated with the temperature sensitivity requirement in the applicable specification sheet. The tachometer generator shall then be excited to reach the stabilized operating temperature of 4.3.4. At this temperature condition the in-phase and quadrature-phase position voltages shall be measured as in 4.8.13 and shall conform to the requirements of 3.6.22. The same test shall be repeated after remaining unexcited for three hours at the high ambient temperature, ± 2°C, associated with the temperature sensitivity requirement in the applicable specification sheet.

4.8.22.5 Zero-speed voltage - temperature sensitivity test. The tachometer generator shall remain unexcited for three hours at the low ambient temperature associated with the temperature sensitivity requirement in the applicable specification sheet. The generator shall then be excited to reach the stabilized operating temperature of 4.3.4. At this temperature condition the fundamental rms null and total rms null voltages shall be measured as in 4.8.12 and shall conform to the requirements of 3.6.22. The same test shall be repeated after remaining unexcited for three hours at the high ambient temperature, ± 2°C, associated with the temperature sensitivity requirement of the applicable specification sheet.

4.8.23 Rotor moment-of-inertia. The rotor moment-of-inertia by torsional oscillations shall be performed as specified in MIL-S-81963 and shall meet the requirements of 3.6.23.

4.8.24 Security of terminals or wire leads. The security of each screw type or solder pin type of terminal or of each wire lead, as applicable to the particular type of tachometer generator, shall be tested in accordance with and shall meet the requirements of MIL-S-81963.

4.9 Environmental tests.

4.9.1 Vibration. Tachometer generators shall be subjected to vibration in accordance with MIL-S-81963 and shall meet the requirements of 3.7.1. During vibration, one-half of the total quantity of units shall be unenergized. The tachometer generators shall be subjected to low impact shock following the vibration requirements.

4.9.2 Shock.

4.9.2.1 Shock, low impact. Tachometer generators shall be subjected to low impact shock in accordance with MIL-S-81963. During the test, the units shall be unenergized and the rotors mechanically free to rotate. The units shall then meet the requirements of 3.7.2.1 and Table VII herein.

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4.9.2.2 Shock, high impact. Tachometer generators shall be subjected to high impact shock in accordance with MIL-S-81963. During the test, the units shall be unenergized and the rotors mechanically free to rotate. The units shall then meet the requirements of 3.7.2.2 and Table VII herein.

4.9.3 Altitude.

4.9.3.1 Altitude, low temperature. The tachometer generators shall be tested in accordance with MIL-S-81963 with the output winding loaded with the standard load specified in the applicable specification sheet. While subjected to this environment, a high impedance voltmeter shall be placed across the output voltage winding and the shaft driven at a speed of 1800 rpm or greater. There shall be no abrupt fluctuations of the output voltage indicative of failure. While still subjected to this environment, the tachometer generators shall conform to the requirements of 3.7.3.1 and Table VII.

4.9.3.2 Altitude, high temperature. The tachometer generators shall be tested in accordance with MIL-S-81963. While subjected to this environment, a high impedance voltmeter shall be placed across the output voltage winding and the shaft driven at a speed of 1800 rpm or greater. There shall be no abrupt fluctuations of the output voltage indicative of failure. While still subjected to this environment, the tachometer generators shall conform to the requirements of 3.7.3.2 and Table VII.

4.9.4 Ambient temperature. The tachometer generator shall be mounted on the standard test fixture of Figure 3 and the output winding loaded with the standard load as specified in 4.3.5.

4.9.4.1 Ambient, low temperature. The unenergized tachometer generator shall be placed in a test chamber and the temperature reduced to -62°C ± 2°C. This temperature shall be maintained for a period of three hours. With the output voltage winding loaded with the standard load, the chamber temperature shall be increased to -55°C ± 2°C and maintained at this temperature for one hour. After this one-hour period, the unit shall be subjected to the dielectric withstanding test of 4.8.5. After successfully withstanding the Table VI test potentials, the energizing winding shall be excited with the rated standard test voltage of 4.3.3, and allowed to attain the stabilized operating temperature of 4.3.4. At this condition and while still in the same environment, the unit shall conform to the requirements of 3.7.4.1.

4.9.4.2 Ambient, high temperature. The unenergized tachometer generator shall be placed in the test chamber and the temperature increased to that high ambient temperature ± 2°C specified in the applicable specification sheet. This temperature shall be maintained for three hours. While still in this same temperature environment, the unit shall be energized and allowed to attain the stabilized operating temperature of 4.3.4. The tachometer generator shall then conform to the requirements of 3.7.4.2.

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4.9.5 Endurance. The tachometer generators shall be mounted on the standard test fixture of Figure 3, and the output winding loaded with the standard load in accordance with 4.3.5. Unless otherwise specified, the 1,000 hour endurance test shall be performed as specified in Table VIII at a speed of 3600 rpm ± 100 rpm at the ambient temperatures, ± 2°C, and at the shaft positions specified. The rotation of the shaft at each shaft position shall be reversed for approximately half the specified time. The tachometer generators shall conform to the requirements of 3.7.5 after completion of this test.

4.9.6 Moisture resistance. The tachometer generators shall be subjected to the moisture resistance test in accordance with MIL-S-81963. One unit shall be unenergized and one unit shall have the reference winding in accordance with 4.3.3. At the end of each 16-hour period in the cycle (completion of step 6), with the humidity controlled at 90 to 95 percent, both units shall be energized with the standard rated voltage for a period of five minutes. This procedure shall be followed during each of the total 10 cycles. After completion of the final cycle, the tachometer generators shall remain unenergized at the standard ambient conditions for 24 hours. The tachometer generators shall subsequently be subjected to and shall meet the requirements of 3.7.6 and Table VII.

4.9.7 Audible noise, structureborne. When required by the applicable specification sheet, structureborne noise test shall be conducted in accordance with MIL-S-81963 while mounted in the test fixture of Figure 6. The unit and fixture combination shall be suspended as shown in Figure 7, in such a manner that the shaft axis and the top edge of the mounting plate are horizontal within 5 degrees. After determination of the prevailing ambient noise level for the broad, one-third octave, and narrow bands with the unit unenergized, the energizing windings shall be energized with the standard test voltage in accordance with 4.3.3. The required vibration levels shall be recorded and shall meet the requirements of 3.7.7.

4.9.8 Salt atmosphere resistance. When required by the applicable specification sheet, tachometer generators shall be subjected to the salt atmosphere test in accordance with MIL-S-81963 and shall meet the requirements of 3.7.8.

4.9.9 Explosion resistance. When required by the applicable specification sheet, tachometer generators shall be subjected to the explosion resistance test in accordance with MIL-S-81963 and shall meet the requirements of 3.7.9.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with MIL-S-81963 unless otherwise specified on the contract (see 6.2 and 6.6).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intended use for tachometer generators covered by this specification must be in accordance with MIL-S-81963.

6.2 Acquisition requirements. Acquisition documents must contain the information specified in MIL-S-81963; and must also specify if a black finish is desired for the housing.

6.3 First article inspection. Information pertaining to first article inspection of tachometer generators should be obtained as specified in MIL-S-81963.

6.4 Rated voltage and frequency tolerances. Normal tolerances for field power supplies are ± 10 percent on voltage and frequency and ± 5 percent on harmonic content. It should be understood that the characteristics specified in the applicable specification sheet are based upon tolerances of ± 1 percent on voltage and frequency and ± 3 percent on harmonic content (see 4.3.3), unless otherwise noted in the applicable specification sheet. Therefore, tachometer generator performance characteristics may be degraded under normal field power supply tolerances.

6.5 Packaging and packing. Criteria for determining the proper level of preservation, packaging and packing shall be as follows:

a. Level A. This level should be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

b. Level B. This level should be used for those items which are to be given multiple domestic shipments with indeterminate period of covered storage.

c. Level C. This level should be used for those items which are to be shipped to domestic installations for immediate use at the first receiving station.

6.6 Definitions.

6.6.1 Tachometer generator. A tachometer generator is an electromechanical device consisting of an energizing winding, an output winding, and a rotor which electromagnetically couples the two windings upon being rotated. When the energizing voltage winding is excited from a single-phase source, the magnitude of the output winding voltage is proportional to the angular velocity of the rotor. The frequency of the output voltage is the same as the fundamental energizing voltage frequency. Ideal characteristics of all tachometer generators would include low inertia rotors, high output-to-null ratios, a strictly linear relationship of output

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voltage as a function of rotor speed over a broad range of ambient temperatures, and a minimum phase shift between the fundamental output and input voltages. The maximum degree to which any one of these desired characteristics is achieved is often at the expense of the remaining. General practice is to group tachometer generators in accordance with their anticipated application.

6.6.1.1 Rate tachometer generator. A rate tachometer generator is characterized by a relatively high output-to-null ratio, and as such, is employed in high-gain rate and computing servomechanism applications. The rotor is of relatively low moment of inertia. Such applications require a high degree of linearity over the range of speeds of the generator.

6.6.1.2 Integrating tachometer generator. An integrating tachometer generator is used in computing applications requiring precise integration of a variable with respect to time. This type of tachometer generator is characterized by very small deviations of the output voltage as a function of temperature and a minimum warm-up time. Temperature control and compensation networks are integral parts of such generators.

6.6.1.3 Damping tachometer generators. Tachometer generators in this group are characterized by high theoretical stall acceleration figures and low null output, and are employed to stabilize the parent system.

6.6.2 Effective resistance. The effective resistance of any winding is equivalent to the DC resistance which, when substituted for the winding being measured, will consume the same power under identical test conditions. It is equal to the impedance of a circuit consisting of a capacitance in parallel with the winding being measured when the capacitor is adjusted for unity power factor or maximum impedance of the circuit.

6.6.3 Speed range. The speed range of a tachometer generator is the specified band of speeds within which performance characteristics are applicable.

6.6.4 Calibration speed. The calibration speed of a tachometer generator is a specified speed within the specified speed range used for standardization and reference purposes. The calibration speed is usually 80 percent of the maximum speed in the speed range.

6.6.5 Output voltage. The output voltage of a tachometer generator is the voltage across the output winding.

6.6.5.1 Speed-sensitive output voltage. The speed-sensitive output voltage is that portion of the output voltage of the fundamental frequency that is a function of speed only. It is numerically equal to one-half the sum of the fundamental output voltages measured at the same speed and test conditions in both directions of rotation as follows:

$$\frac{V_{iccw} + V_{icw}}{2}$$

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Where: V is the magnitude of the fundamental component of the output voltage.

6.6.5.1.1 In-phase speed-sensitive output voltage. The in-phase speed-sensitive voltage is that component of the speed-sensitive output voltage that has a phase which is the same or 180 degrees from the reference phase. It is equal to one-half the algebraic difference between the in-phase speed-sensitive output voltages measured at the same speed and test conditions in both directions of rotation as follows:

$$\frac{Viccw - Vicw}{2}$$

Where: Vi is the in-phase speed-sensitive output voltage.

6.6.5.1.2 Quadrature-phase speed-sensitive output voltage. The quadrature-phase speed-sensitive output voltage is that component of the speed-sensitive output voltage that has a phase which is 90 degrees or 270 degrees from the reference phase. It is equal to one-half the algebraic difference between the quadrature-phase speed-sensitive output voltages measured at the same speed and test conditions in both directions of rotation as follows:

$$\frac{Vqccw - Vqcw}{2}$$

Where: Vq is the quadrature-phase speed-sensitive output voltage.

6.6.5.2 Zero-speed output voltage. The zero-speed output voltage of a tachometer generator is the voltage present across the output winding that is a function of rotor position only.

6.6.5.2.1 Total rms null voltage. The total rms null voltage of a tachometer generator is the magnitude of the total rms zero-speed output voltage at that rotor position which produces a maximum value.

6.6.5.2.2 Fundamental rms null voltage. The fundamental rms null voltage of a tachometer generator is the magnitude of the rms zero-speed output voltage of fundamental frequency at that rotor position which produces a maximum value.

6.6.5.2.3 In-phase null voltage. The in-phase null voltage of a tachometer generator is the magnitude of that component of rms zero-speed output voltage that has a phase which is the same or 180 degrees from the reference phase and is measured at a rotor position which produces a maximum value for the in-phase null voltage.

6.6.5.2.4 Quadrature-phase null voltage. The quadrature-phase null voltage of a tachometer generator is the magnitude of that component of rms zero-speed output voltage that has a phase which is 90 degrees from the

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reference phase and is measured at a rotor position which produces a maximum value for the quadrature-phase null voltage.

6.6.5.3 Position error voltage. The position error voltage of a tachometer generator is one-half the difference between the maximum and minimum values of that component of the fundamental frequency zero-speed rms output voltage as this varies with the angular position of the rotor between 0 and 360 degrees.

6.6.5.3.1 In-phase position error voltage. In-phase position error voltage of a tachometer generator is one-half the algebraic difference between the maximum and minimum values of that component of the fundamental frequency zero-speed rms output voltage, as this varies with the angular position of the rotor between 0 and 360 degrees, which is in phase with or 180 degrees out of phase from the reference phase. (See Figure 5)

6.6.5.3.2 Quadrature-phase position error voltage. The quadrature-phase position error voltage of a tachometer generator is one-half the algebraic difference between the maximum and minimum values of that component of the fundamental frequency rms zero-speed output voltage, as this varies with the angular position of the rotor between 0 and 360 degrees, which has a phase displacement of 90 degrees from the reference phase. (See Figure 5)

6.6.5.4 Axis error voltage. The axis error voltage of a tachometer generator is that portion of the output voltage of fundamental frequency that is neither a function of rotor position nor rotor speed.

$$\frac{Vccw + Vcw}{2}$$

Where: Vi is the magnitude of the fundamental component of the output voltage.

6.6.5.4.1 In-phase axis error voltage. The in-phase axis error voltage of a tachometer generator is the component of the axis error voltage that has a phase which is the same or 180 degrees from the reference phase. It is equal to one-half the algebraic sum of the in-phase speed-sensitive output voltages measured at the same speed and test conditions in both directions of rotation as follows:

$$\frac{Viccw + Vicw}{2}$$

Where: Vi is the in-phase speed-sensitive output voltage. In terms of the in-phase (zero speed output voltage) rotor position error voltage, refer to Figure 4.

6.6.5.4.2 Quadrature-phase axis error voltage. The quadrature-phase axis error voltage of a tachometer generator is that component of the axis error voltage that has a phase which is 90 degrees or 270 degrees from the

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reference time phase. It is equal to one-half the algebraic sum of the quadrature-phase speed-sensitive output voltages measured at the same speed and test conditions in both directions of rotation as follows:

$$\frac{Vqccw + Vqcw}{2}$$

Where: Vq is the quadrature-phase speed-sensitive output voltage. In terms of the quadrature-phase (zero speed output voltage) rotor position error voltage, refer to Figure 4.

6.6.5.5 Speed-sensitive transformation ratio. The speed-sensitive transformation ratio of a tachometer generator is the ratio of the speed-sensitive output voltage to the fundamental energizing voltage at any specified speed in the speed range, usually the calibration speed.

6.6.5.5.1 In-phase speed-sensitive transformation ratio. The in-phase speed-sensitive transformation ratio (3.6.14) of a tachometer generator is the ratio of the in-phase speed sensitive output voltage to the fundamental energizing voltage at the specified calibration speed or speeds.

6.6.5.5.2 Quadrature-phase speed-sensitive transformation ratio. The quadrature-phase speed-sensitive transformation ratio of a tachometer generator is the ratio of the quadrature-phase speed-sensitive output voltage to the fundamental energizing voltage at any specified speed in the speed range, usually the calibration speed.

6.6.6 Output gradient. The output gradient of a tachometer generator is the approximate slope of the curve of the speed-sensitive output voltage as a function of speed. It is expressed in volts per 1000 rpm at any specified speed in the speed range.

6.6.7 Phase shift. The phase shift of an integrating tachometer generator is the difference in time phase between the fundamental energizing winding voltage and the output voltage (6.6.5). It is equal to the arc tangent of the ratio of the quadrature-phase speed sensitive output voltage to the in-phase speed-sensitive output voltage in degrees, at the same speed and conditions of test.

6.6.8 Linearity. The linearity of a tachometer generator is a ratio of the difference between the measured output voltage (E_o), and the calculated output voltage (E_{oc}), to the calculated output voltage at maximum speed (E_{max}), expressed in percent.

$$E_{oc} = \frac{N_o}{N_c} \quad (E_c)$$

$$E_{max} = \frac{N_{max}}{N_c} \quad (E_c)$$

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Where:

N_o = any speed in the speed range

N_c = the calibration speed

N_{max} = the maximum speed in the speed range

E_o = the output voltage measured at speed, N

E_c = the output voltage measured at the calibration speed, N_c

Then:

$$\text{Linearity} = \frac{E_o - E_{oc}}{E_{max}} \ (100\%)$$

6.6.8.1 In-phase linearity. The in-phase linearity, expressed in percent, is defined as:

$$\text{In-phase linearity} = \left[\frac{ITR_o - \frac{N_o}{N_c}(ITR_c)}{\frac{N_{max}}{N_c} - (ITR_c)} \right] (100\%)$$

Where: ITR_o is the in-phase speed-sensitive transformation ratio at any speed (N_o), in the specified speed range.

ITR_c is the in-phase speed-sensitive transformation ratio at the calibration speed (N_c).

6.6.9 Rated voltage. The rated voltage of a tachometer generator is that voltage specified in the applicable specification sheet for the energizing winding.

6.6.10 Energizing winding. The energizing winding of a tachometer generator is that winding which is excited with a specified fixed voltage and frequency.

6.6.11 Output winding. The output winding of a tachometer generator is that winding whose voltage output is proportional to the angular velocity of the rotor when the energizing winding is excited. The frequency of the output voltage is the rated frequency.

6.6.12 Reference phase. The reference phase of a tachometer generator is the phase of the fundamental voltage applied to the energizing winding and is regarded as being zero degrees.

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6.6.13 Temperature rise. The temperature rise of a tachometer generator is the increase of the internal temperature of the unit above the ambient temperature due to the dissipation of excitation power and heater element power.

6.6.14 Warm-up time. The warm-up time of a tachometer generator is the time required for the unit to meet performance requirements after being stored for three hours at a specified ambient temperature.

6.6.15 Voltage sensitivity. The voltage sensitivity of a tachometer generator is the change in the electrical or mechanical characteristics due to prescribed deviations from the rated value of the standard voltage or energizing voltage.

6.6.16 Frequency sensitivity. The frequency sensitivity of a tachometer generator is the change in the electrical or mechanical characteristics due to prescribed deviations from the rated value of the fundamental frequency.

6.6.17 Temperature sensitivity. The temperature sensitivity of a tachometer generator is the change in the electrical or mechanical characteristics due to a prescribed deviation in the ambient temperature from 23°C ± 2°C.

6.6.18 Shaft radial play. The shaft radial play of a tachometer generator rotor is the shaft displacement perpendicular to the shaft axis due to reversal of a force applied perpendicular to the shaft axis.

6.6.19 Shaft end play. The shaft end play of a tachometer generator rotor is the axial shaft displacement due to reversal of a force applied axially to the shaft.

6.6.20 Units of measurement. Unless otherwise specified, the units of measurement are as follows:

Angles	=	degrees and minutes
Potential	=	volts rms
Impedance	=	ohms
Current	=	amperes rms
Power	=	watts
Temperature	=	degrees Centigrade
Torque	=	ounce-inches
Angular velocity	=	revolutions per minute (rpm)

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6.7 Subject term (key word) listing.

Axis error voltage
Damping tachometer generators
Electromechanical device
First article
Frequency sensitivity
Heater compensated
In-phase linearity
In-phase transformation ratio
Integrating tachometer generator
Network compensated
Output winding
Position error voltage
Quadrature-phase
Qualification
Qualified products list (QPL)
Quality conformance
Rate tachometer generators
Servocomponent
Specification
Speed-sensitive transformation ratio
Temperature sensitivity
Voltage sensitivity
Warm-up time
Zero speed voltage

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.9 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreements reached by the NATO Working Group on Analogue and Digital Servocomponents (AC/301(SG/A)(WG/5). When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.10 Applicable international documentation. NATO documents applicable to this specification are Allied Standard Publication (AStanP)-3, Volume 6680 Chapter 2, NATO Electronic/Electrical Preferred Parts List, Tachometer Generators, AC; AStanP-4, Volume 6680 Chapter 2, NATO Electronic/Electrical Technical Recommendation, Tachometer Generators, AC; and AStanP-5, Volume 6680 Chapter 2, NATO Quality Assessment Recommendation for Electronic/Electrical Parts, Tachometer Generators, AC.

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TABLE I. Nomenclature

Tachometer Generator	15	IH	4	A
Item Name	Size	Function	Supply Frequency	Modification
(1.2.1)	(1.2.1.1)	(1.2.1.2)	(1.2.1.3)	(1.2.1.4)

TABLE II. Example of part identifying number.

M	22820/1	-01	C
Military Designation	Specification Sheet Number	Dash No.	Latest Modification Letter

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TABLE III. Standard dimensions for tachometer generators.

(See Figure 2 for location of lettered dimensions)

Dimension 1/	A	B 2/	C	D	E
Tolerance	±0.010	±0.010	+0.003 -0.005	+0.0000 -0.0005	+0.0000 -0.0005
Tachometer Generator Size 15	0.540	0.312	1.437	1.312	0.8750

SPUR GEAR DATA F				Pitch Diameter	Outside Diameter	Root Diameter
Tolerance	No. of Teeth	Diametral Pitch	Pressure Angle	+0.000 -0.001	+0.0000 -0.0005	Maximum
Tachometer Generator Size 15	21	120	20°	0.185	0.1872	0.155

Dimension	G	H	I	J	K
Tolerance	As specified in the applicable specification sheet.	±0.005	±0.005	±0.005	±0.005
Tachometer Generator Size 15		0.040	0.132	0.093	0.078

Dimension	L	M	N	O	R 3/
Tolerance	±.005	±.005	±.005	±.003	±.005
Tachometer Generator Size 15	1.312	.140	.200	1.100	1.031

1/ Unless specified otherwise, dimensions are in inches.
2/ B is length for full depth of tooth.
3/ R is applicable for terminal blocks only.

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TABLE IV. Termination identification marking.

Terminal No.	Wire Lead Color	Winding
1	Orange	Energizing
3	Gray	Energizing
4	Brown	Output
2	Blue	Output

TABLE V. Terminal hardware.

Tachometer Generator Type	Hardware	Military Standard	Number Required
15IH4A	Machine Screw	MS35276-212	4 each 1/
	Lock Washer	MS35338-135	4 each 1/
15RG4A	Machine Screw	MS35276-212	4 each 1/
	Lock Washer	MS35338-135	4 each 1/

1/ Units with 6 terminals will require 6 each machine screws and lock washers.

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TABLE VI. Test voltages and application points.

Excitation Voltage	1 to Frame	2 to Frame	1 to 2	Heater Winding	
				5 to Frame	5 to 1 5 to 2
Under 60 Volts	385 to 400	385 to 400	385 to 400	385 to 400	385 to 400
Over 60 Volts	720 to 750	720 to 750	720 to 750	720 to 750	720 to 750

NOTE:

1. Subsequent tests shall be performed at 80 percent of the initial voltage.

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TABLE VII. First article and quality conformance inspection tests.

Test No.	Requirement	Test	Test method for examination	Inspection	
				First article Sample Number	Quality Conformance
1	3.6.1	4.8.1	Visual and mechanical inspection	1, 2, 3, 4	X
2	3.6.2	4.8.2	Mechanical breakaway torque	1, 2, 3, 4	X
3	3.6.3	4.8.3	Shaft radial and end play	1, 2, 3, 4	X
4	3.6.4	4.8.4	Total shaft runout (smooth portion)	1, 2, 3, 4	X
5	3.6.5	4.8.5	Dielectric withstanding voltage	1, 2, 3, 4	X
6	3.6.6	4.8.6	Insulation resistance	1, 2, 3, 4	X
7	3.6.7	4.8.7	Current and power, heater windings	1, 2, 3, 4	X
8	3.6.8	4.8.8	Current	1, 2, 3, 4	X
9	3.6.9	4.8.9	Power	1, 2, 3, 4	X
10	3.6.10	4.8.10	Polarity	1, 2, 3, 4	X
11	3.6.11	4.8.11	Axis-error voltage	1, 2, 3, 4	X
12	3.6.12	4.8.12	Zero-speed output voltage	1, 2, 3, 4	X
13	3.6.13	4.8.13	Position error voltage	1, 2, 3, 4	X
14	3.6.14	4.8.14	In-phase speed-sensitive transformation ratio	1, 2, 3, 4	X
15	3.6.15	4.8.15	Phase shift	1, 2, 3, 4	X

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TABLE VII. First article and quality conformance inspection tests - (Continued).

Test No.	Requirement	Test	Test method for examination	Inspection	
				First article Sample Number	Quality Conformance
16	3.6.16	4.8.16	Warm-up time	1, 2, 3, 4	-
17	3.6.17	4.8.17	Impedance	1, 2, 3, 4	-
18	3.6.18	4.8.18	In-phase linearity	1, 2, 3, 4	-
19	3.6.19	4.8.19	Voltage sensitivity during test nos. 11, 12, 13, 14, and 15	1, 2, 3, 4	-
20	3.6.20	4.8.20	Frequency sensitivity during test nos. 11, 12, 13, 14, and 15	1, 2, 3, 4	-
21	3.6.21	4.8.21	Temperature rise	1, 2, 3, 4	-
22	3.6.22	4.8.22	Temperature sensitivity during test nos. 11, 12, 13, 14, 15, and 18	1, 2, 3, 4	-
23	3.6.23	4.8.23	Rotor moment of inertia	1, 2 3, 4	-
24	3.6.24	4.8.24	Security of terminals or wire leads	1, 2, 3, 4	-
25	3.7.1	4.9.1	Vibration, followed by test nos. 2, 3, 5, 6, 11, 12, 13, 14, 15, and 18	1, 2, 3, 4	-
26	3.7.2.1	4.9.2.1	Shock, low impact, followed by test nos. 2, 3, 5, 6, 11, 12, 13, 14, 15, and 18	1, 2, 3, 4	-

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TABLE VII. First article and quality conformance inspection tests - (Continued).

Test No.	Requirement	Test	Test method for examination	Inspection	
				First article Sample Number	Quality Conformance
27	3.7.2.2	4.9.2.2	Shock, high impact, followed by test nos. 2, 3, 5, 6, 11, 12, 13, 14, 15, and 18	1, 2, 3, 4	-
28	3.7.3.1	4.9.3.1	Altitude, low temperature, during test no. 6	1, 2	-
29	3.7.3.2	4.9.3.2	Altitude, high temperature, during test no. 6	1, 2	-
30	3.7.4.1	4.9.4.1	Ambient low temperature, during test nos. 5 and 6	3, 4	-
31	3.7.4.2	4.9.4.2	Ambient high temperature, during test nos. 5 and 6	3, 4	-
32	3.7.5	4.9.5	Endurance, followed by tests 2, 3, 5, 6, 11, 12, 13, 14, 15 and 18	1, 2	-
33	3.7.6	4.9.6	Moisture resistance, followed by test nos. 2, 5, 6, 11, 12, 13, 14, 15, and 18	3, 4	-
34	3.7.7	4.9.7	Audible noise structureborne 1/	1, 2, 3, 4	-
35	3.7.8	4.9.8	Salt atmosphere resistance 1/	1, 2	-
36	3.7.9	4.9.9	Explosion resistance 1/	3, 4	-

1/ Test Nos. 34, Audible noise structureborne; 35, Salt atmosphere; and 36, Explosion resistance shall be performed only when required by the specification sheet.

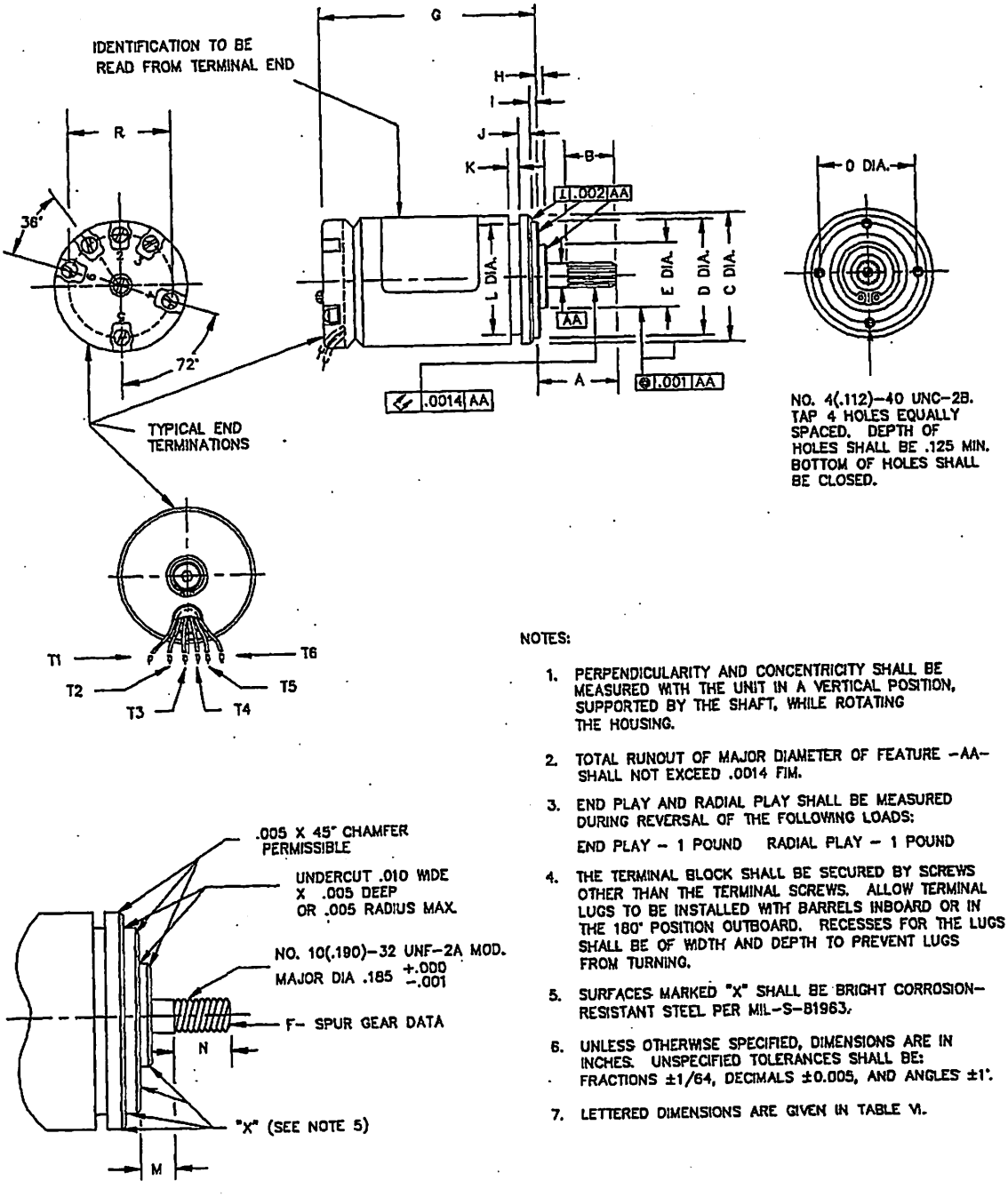
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TABLE VIII. Endurance test time schedule.

Time (hours)	Temperature °C (± 5 degrees)	Shaft Position
64 (±2)	-55 degrees	Horizontal
24 (±1)	1/	Vertical Up
24 (±1)	1/	45 Degrees Up
24 (±1)	1/	45 Degrees Down
24 (±1)	1/	Vertical Down
840 (±7)	25 degrees	Horizontal

1/ Indicates that high ambient temperature specified in the applicable specification sheet.

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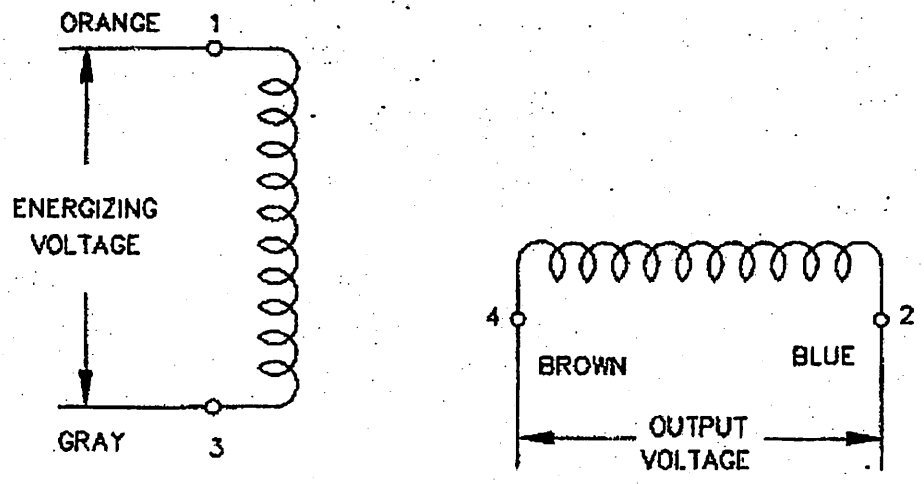


NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION, SUPPORTED BY THE SHAFT, WHILE ROTATING THE HOUSING.
2. TOTAL RUNOUT OF MAJOR DIAMETER OF FEATURE -AA- SHALL NOT EXCEED .0014 FIM.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING REVERSAL OF THE FOLLOWING LOADS:
   END PLAY - 1 POUND RADIAL PLAY - 1 POUND
4. THE TERMINAL BLOCK SHALL BE SECURED BY SCREWS OTHER THAN THE TERMINAL SCREWS. ALLOW TERMINAL LUGS TO BE INSTALLED WITH BARRELS INBOARD OR IN THE 180° POSITION OUTBOARD. RECESSES FOR THE LUGS SHALL BE OF WIDTH AND DEPTH TO PREVENT LUGS FROM TURNING.
5. SURFACES MARKED "X" SHALL BE BRIGHT CORROSION-RESISTANT STEEL PER MIL-S-81963.
6. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES. UNSPECIFIED TOLERANCES SHALL BE: FRACTIONS ±1/64, DECIMALS ±0.005, AND ANGLES ±1°.
7. LETTERED DIMENSIONS ARE GIVEN IN TABLE VI.

FIGURE 1. Outline drawing for size 15 tachometer generator.

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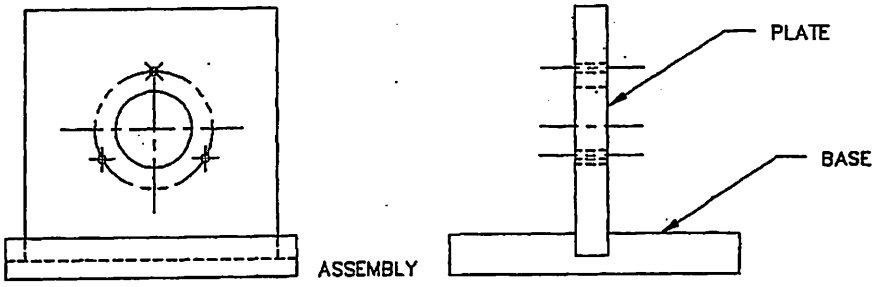
NOTE:

1. WITH THE ROTOR TURNING IN A CCW DIRECTION THE VOLTAGE AT 2 SHALL BE 180° OUT OF PHASE WITH THE FUNDAMENTAL VOLTAGE AT 1.

   TERMINALS 3 AND 4 (OR WIRE LEADS GRAY AND BROWN) ARE STRAPPED AND GROUNDED TO FRAME FOR THE ABOVE TEST.

FIGURE 2. Standard tachometer generator schematic and test connection diagram.

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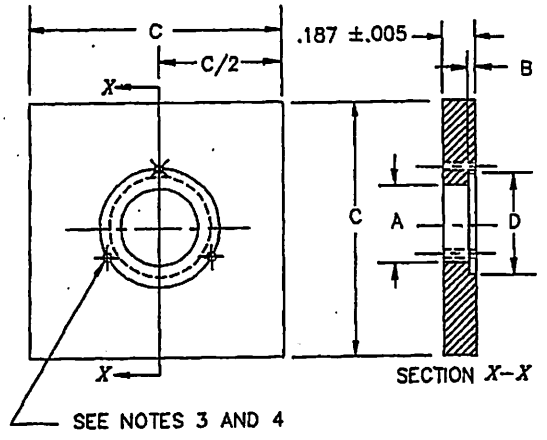
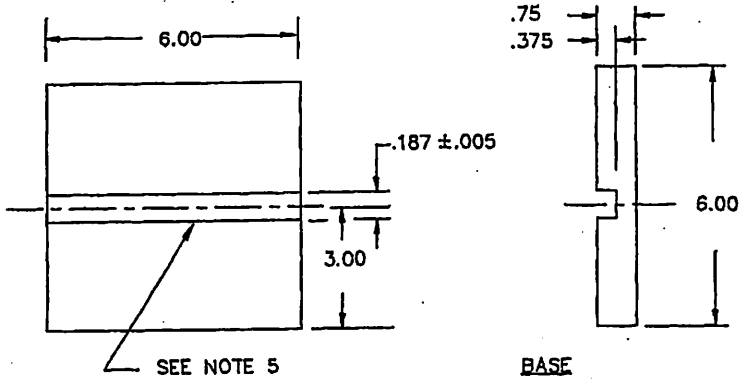


TABLE OF LETTERED DIMENSIONS				
SIZE	A	B	C	D
05	0.376	----	2.214	----
08	0.501	----	2.250	----
11	0.626	0.057	3.200	1.005
15	0.876	0.127	4.300	1.317
18	0.938	0.127	5.250	1.567

PLATE

MATERIAL: ALUMINUM ALLOY, QQ-A-250/4
FINISH: ANODIZE PER MIL-A-8625, TYPE II, CLASS 2 BLACK



BASE

MATERIAL: LAMINATED PHENOLIC

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE: 2 PLACE DECIMALS ±.02
   3 PLACE DECIMALS $^{+.010}_{-.000}$
3. THREE THREADED HOLES EQUALLY SPACED ON A DIAMETER APPROPRIATE FOR THE MOUNTING HARDWARE USED.
4. MOUNTING HARDWARE CLAMP ASSEMBLY PER MS17183, OR EQUIVALENT, DIMENSIONED FOR THE PARTICULAR SERVOCOMPONENT.
5. SLOT IN BASE SHALL BE LIGHT PRESS FIT WITH PLATE.

FIGURE 3. Standard test fixture for size 18 and smaller tachometer generators.

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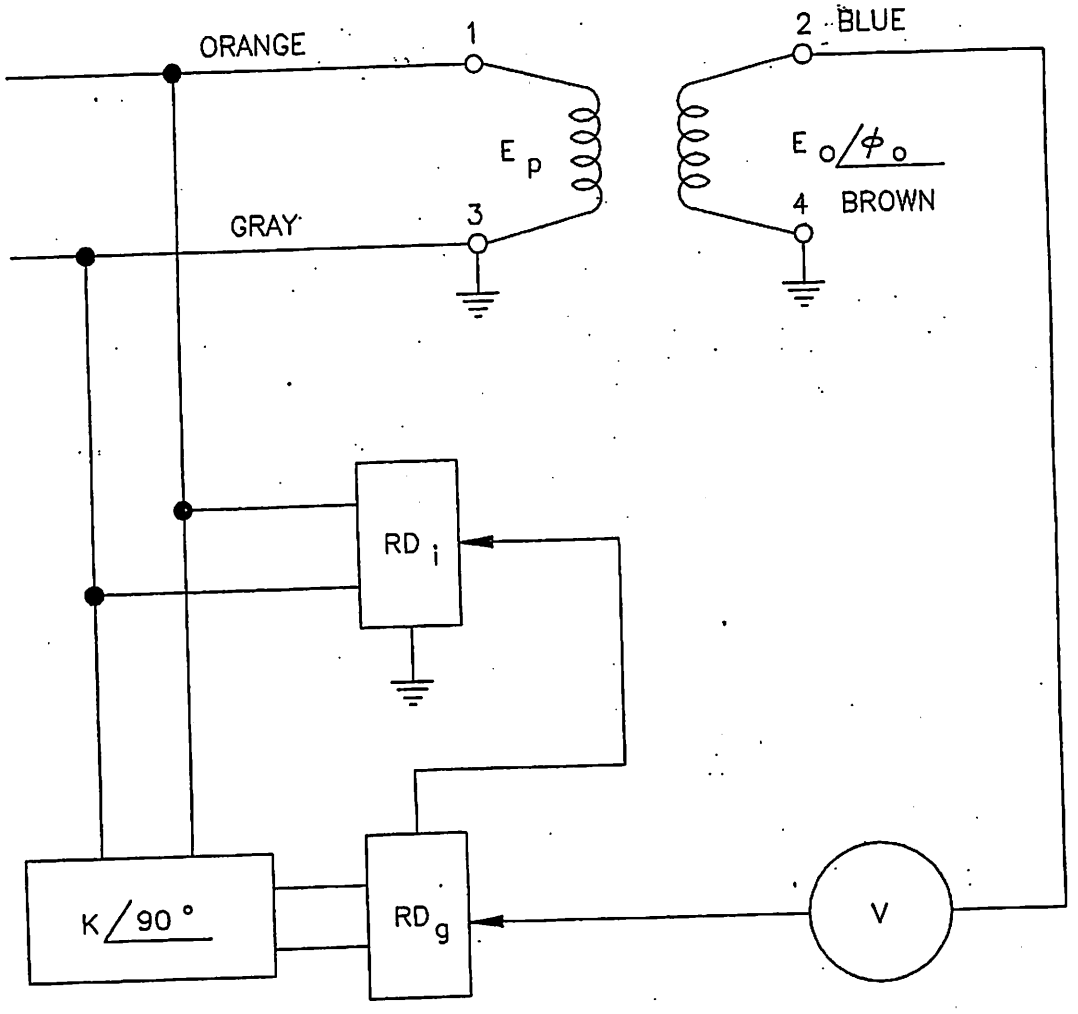


FIGURE 4. Speed-sensitive output measuring equipment.

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NOTES: (FIGURE 4)

Where:

1. RD_i is a ratio device which the indicated ratio associated with the lowest measured specified speed in the speed range shall be accurate to within ±0.007 percent for integrating tachometers; shall be accurate to within ±0.1 percent for rate tachometers; and shall be accurate to within ±0.5 percent for damping tachometers.

2. RD_g is the same as RD_i.

3. K±90° is a quadrature-generating device that shifts the time-phase of E_p 90 degrees ±1 degree, and which must be maintained to within 0.25 degree during linearity tests.

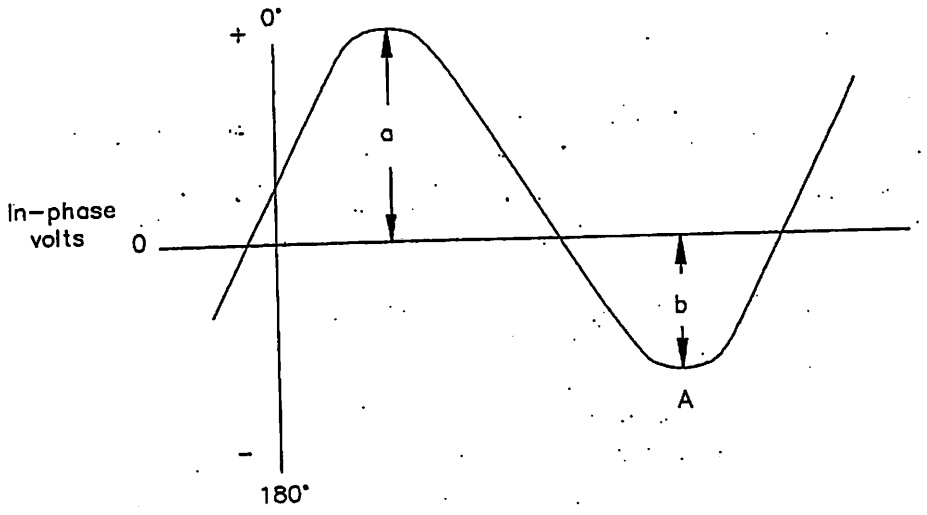
4. K is the gain of the device and must be known and maintained to within 0.01 percent.

5. V is a high impedance voltmeter of the type which is tuned to the fundamental frequency. The response of the filter shall be such that all frequencies above twice the fundamental and frequencies below one-half the fundamental shall be attenuated at least 40 decibels. V may also be a high-impedance phase-sensitive type voltmeter and possess characteristics such that the voltmeter quadrature-detecting axis is within 89 to 91 electrical degrees of the reference time axis. The signal input shall be filtered at the fundamental frequency, and the filter shall possess the characteristic that the response at frequencies greater than twice the fundamental be attentuated at least 40 decibels. The maximum full scale sensitivity shall be at least 1 millivolt.

6. E_o is the fundamental speed-sensitive output voltage of the tachometer under test.

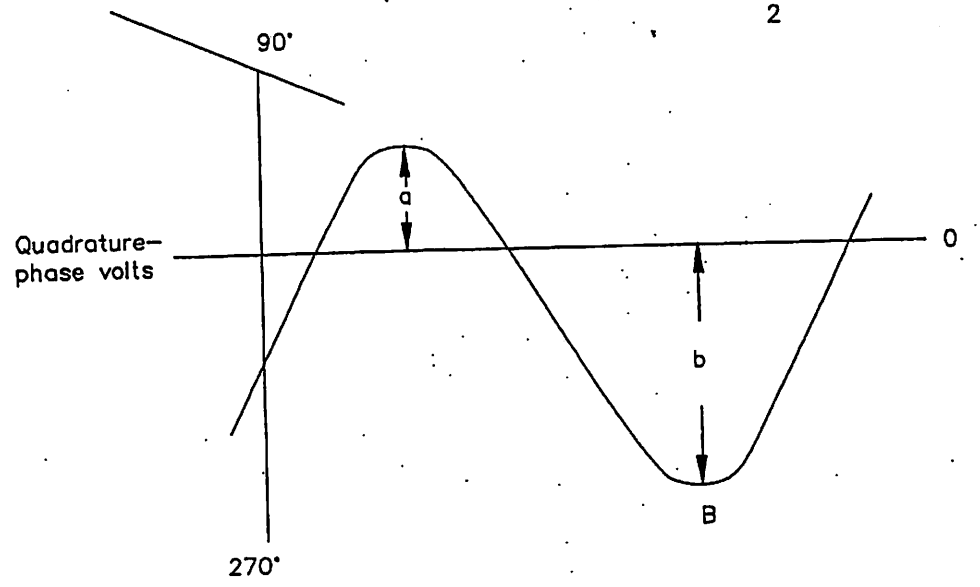
7. ϕ_o is the electrical time-phase of the output voltage, E_o, with respect to the time-phase of the primary energizing voltage, E_p. ϕ_o is the phase shift of the tachometer by definition.

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$$\text{In-phase position error voltage} = \frac{a-(-b)}{2}$$

$$\text{In-phase axis error voltage} = \frac{a+(-b)}{2}$$



$$\text{Quadrature-phase position error voltage} = \frac{a-(-b)}{2}$$

$$\text{Quadrature-phase axis error voltage} = \frac{a+(-b)}{2}$$

FIGURE 5. Position voltages.

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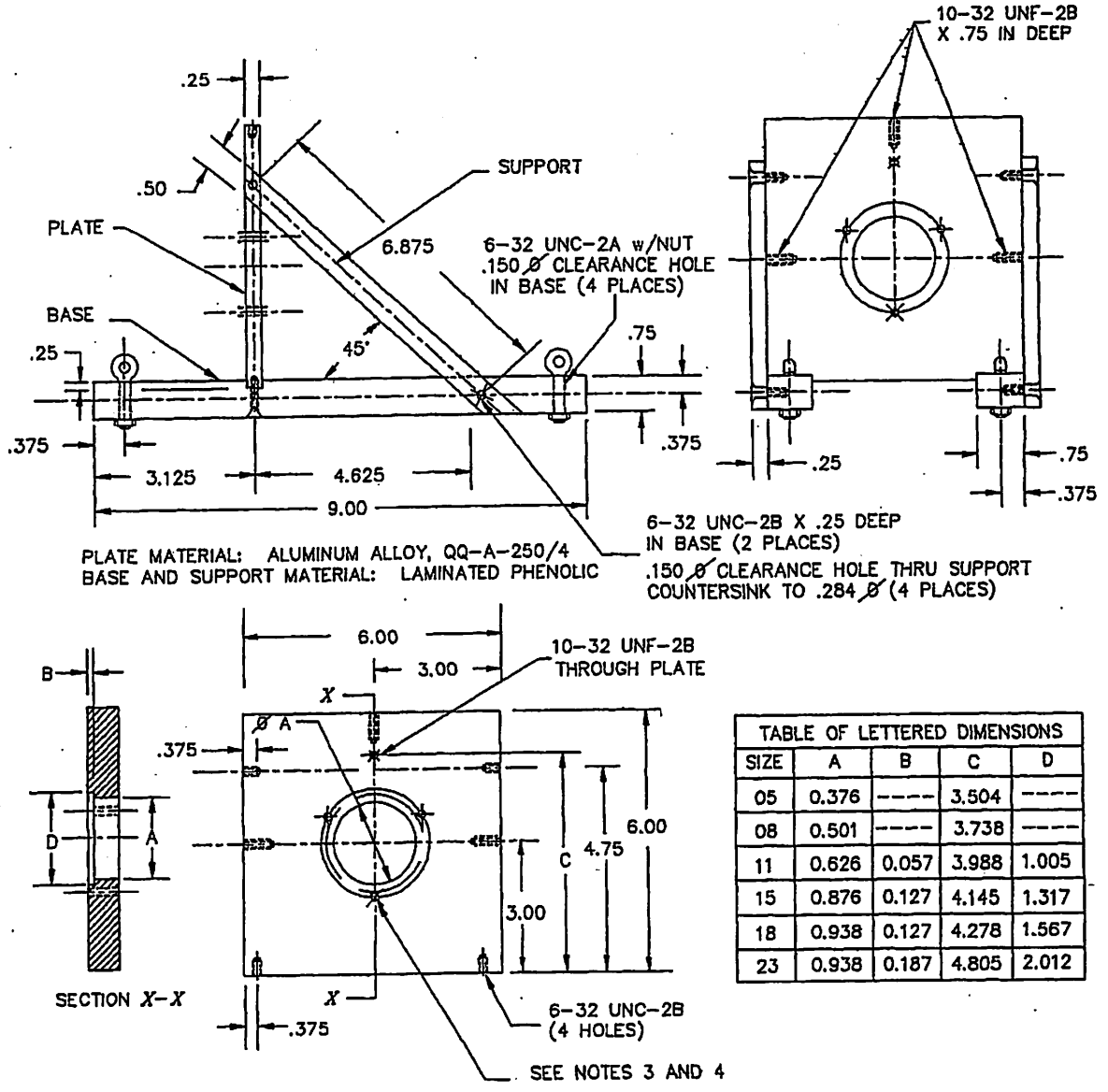


TABLE OF LETTERED DIMENSIONS				
SIZE	A	B	C	D
05	0.376	----	3.504	----
08	0.501	----	3.738	----
11	0.626	0.057	3.988	1.005
15	0.876	0.127	4.145	1.317
18	0.938	0.127	4.278	1.567
23	0.938	0.187	4.805	2.012

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE: 2 PLACE DECIMALS ±.02
   3 PLACE DECIMALS +.010
   -.000
3. THREE THREADED HOLES EQUALLY SPACED ON A DIAMETER APPROPRIATE FOR THE MOUNTING HARDWARE USED.
4. MOUNTING HARDWARE CLAMP ASSEMBLY PER MS17183, OR EQUIVALENT, DIMENSIONED FOR THE PARTICULAR SERVOCOMPONENT.

FIGURE 6. Tachometer generator structureborne noise test fixture.

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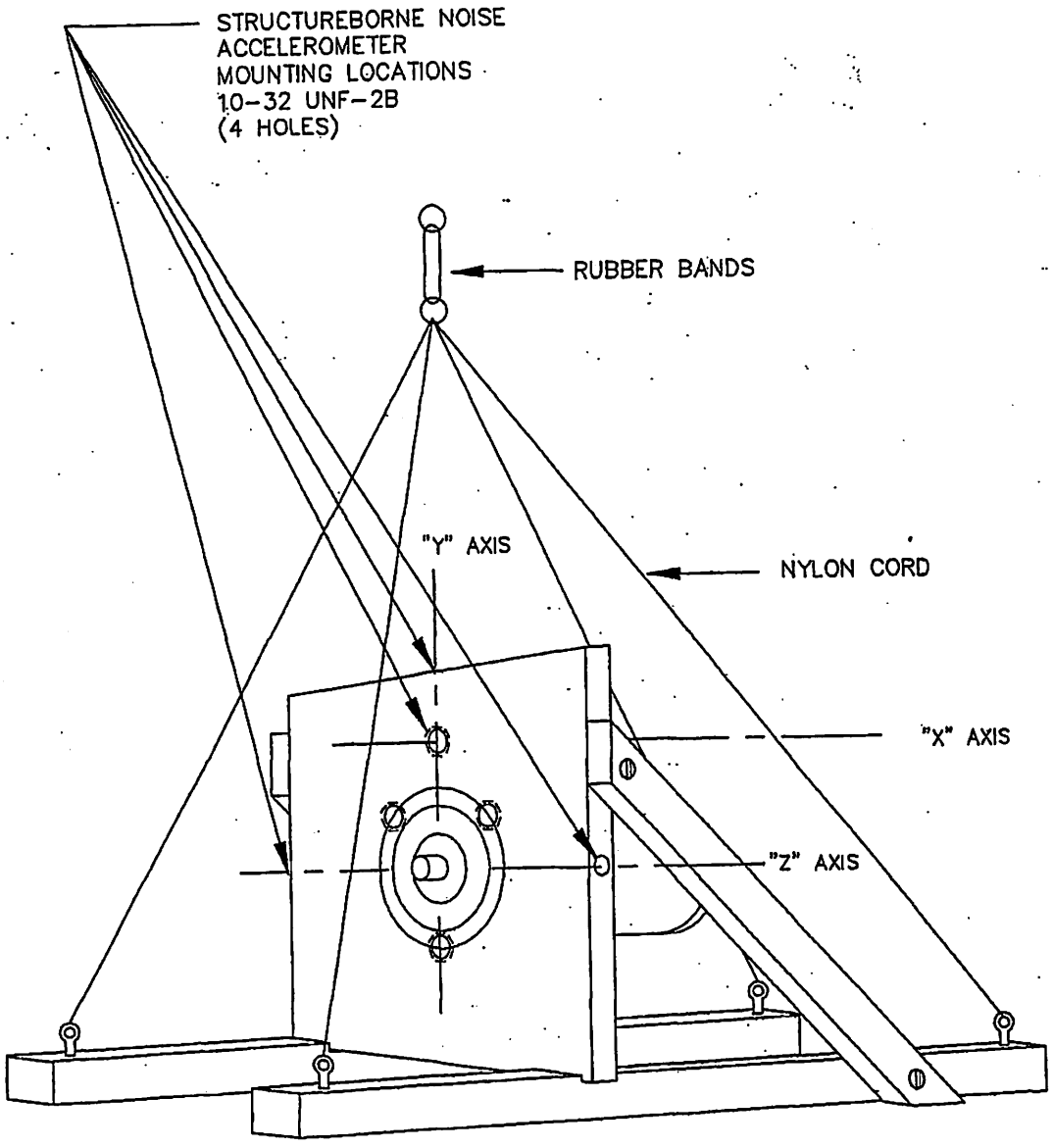


FIGURE 7. Test fixture application for structureborne noise test.

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